

AthenaK output for post-processing analysis

This document collects which AthenaK output variables are needed to perform a certain analysis in post-processing, e.g. the visualization of the evolution of kinetic and magnetic energy in the case of GRMHD simulations.

Ejecta and Poynting flux

VARIABLES

Metric: `adm_gxx`, `adm_gxy`, `adm_gxz`, `adm_gyy`, `adm_gyz`, `adm_gzz`

Primitives: `dens`, `s00`, `temperature`, `velx`, `vely`, `velz`, `bcc1`, `bcc2`, `bcc3`

Gauge: `z4c_alpha`, `z4c_betax`, `z4c_betay`, `z4c_betaz`

Total: $N_{\text{vars}} = 23$

OUTPUT TYPE

```
<output1>
file_type = sph
variable = ... # see variable list
dt = 20.0
radius = 400.0 # radii = 300.0 400.0 ...
```

SETUP

Output time: 10M or 20M (the latter might already be too large)

Resolution: $(\theta, \phi)_{\text{low}} = (128, 256)$ or $(\theta, \phi)_{\text{high}} = (256, 512)$

Radii: Between 300M and 500M

DISK USAGE

- Low resolution - $N_{\text{vars}} \times N_{\theta} \times N_{\phi} \times 8 \text{ bytes} = 23 \times 128 \times 256 \times 8 \approx 6.03\text{MB}(r\Delta t)^{-1}$
- High resolution - $N_{\text{vars}} \times N_{\theta} \times N_{\phi} \times 8 \text{ bytes} = 23 \times 256 \times 512 \times 8 \approx 24.12\text{MB}(r\Delta t)^{-1}$

	10M	20M
• Total (100ms or 20000M):		
	Low resolution	High resolution
	$\approx 12.06 \text{ GBr}^{-1}$	$\approx 48.24 \text{ GBr}^{-1}$
	$\approx 6.03 \text{ GBr}^{-1}$	$\approx 24.12 \text{ GBr}^{-1}$

NOTE To save disk-usage, this can be paired with the output from **Neutrino luminosity and average energy density** and also **GRMHD butterfly diagrams**.

Neutrino luminosity and average energy density

VARIABLES

M1 variables: `rad_m1_F`, `rad_m1_E`, `rad_m1_N`

Metric: `adm_gxx`, `adm_gxy`, `adm_gxz`, `adm_gyy`, `adm_gyz`, `adm_gzz`

Gauge: `z4c_alpha`, `z4c_betax`, `z4c_betay`, `z4c_betaz`

Total: $N_{\text{vars}} = N_{\text{vars,E/N}} \times N_{\text{species}} + \underbrace{N_{\text{vars,F}}}_{3 \text{ vector components}} \times N_{\text{species}} + N_{\text{vars,metric/gauge}} = 2 \times 4 + 3 \times 4 + 10 = 30$

OUTPUT TYPE

```
<output2>
file_type = sph
variable = ... # see variable list
dt = 20.0
radius = 400.0 # radii = 300.0 400.0 ...
```

■ SETUP

Output time: 10M or 20M (the latter might already be too large)

Resolution: $(\theta, \phi)_{\text{low}} = (128, 256)$ or $(\theta, \phi)_{\text{high}} = (256, 512)$

Radii: Between 300M and 500M

■ DISK USAGE

- Low resolution - $N_{\text{vars}} \times N_{\theta} \times N_{\phi} \times 8 \text{ bytes} = 30 \times 128 \times 256 \times 8 \approx \boxed{7.86\text{MB}(r\Delta t)^{-1}}$
- High resolution - $N_{\text{vars}} \times N_{\theta} \times N_{\phi} \times 8 \text{ bytes} = 30 \times 256 \times 512 \times 8 \approx \boxed{31.46\text{MB}(r\Delta t)^{-1}}$

	10M	20M
• Total (100ms or 20000M):		
	Low resolution	Low resolution
	$\approx 15.72 \text{ GBr}^{-1}$	$\approx 7.86 \text{ GBr}^{-1}$
	High resolution	High resolution
	$\approx 62.92 \text{ GBr}^{-1}$	$\approx 31.46 \text{ GBr}^{-1}$

■ **NOTE** To save disk-usage, this can be paired with the output from **Ejecta** and also **GRMHD butterfly diagrams**.

Tracers

■ VARIABLES

Derived variables: `u_t`

M1 variables (fluid frame): `rad_m1_J`, `rad_m1_H`, `rad_m1_chi`, `rad_m1_fnu`, `rad_m1_n`

Primitives: `velx`, `vety`, `velz`, `dens`, `temperature`, `s_00`

Gauge: `z4c_alpha`, `z4c_betax`, `z4c_betay`, `z4c_betaz`

Total: $\boxed{N_{\text{vars}}} = N_{\text{vars},J/n/\chi} \times N_{\text{species}} + \underbrace{N_{\text{vars},H}}_{3 \text{ vector components}} \times N_{\text{species}} + \underbrace{N_{\text{vars},\text{fnu}}}_{4 \text{ vector components}} \times N_{\text{species}} + N_{\text{vars},\text{der./primitives/gauge}} =$

$$3 \times 4 + 3 \times 4 + 4 \times 4 + 11 = \boxed{51}$$

■ OUTPUT TYPE

```
<output3>
file_type = sph
variable = ... # see variable list
dt = 20.0
r_min = 150.0
r_max = 1000.0
r_spacing = linear
nradii = 128
ntheta = 128
```

■ **SETUP** Output time: 20M might be too expensive, but 50M might already be too coarse Resolution: $(r, \theta, \phi)_{\text{low}} = (128, 128, 256)$ or $(r, \theta, \phi)_{\text{high}} = (128, 256, 512)$

Radii: 128 radii between the outer part of the disk and the domain is a good choice

■ DISK USAGE

- Low resolution - $N_{\text{vars}} \times N_r \times N_{\theta} \times N_{\phi} \times 8 \text{ bytes} = 51 \times 128 \times 128 \times 256 \times 8 \approx \boxed{213.91\text{MB}(\Delta t)^{-1}}$
- High resolution - $N_{\text{vars}} \times N_r \times N_{\theta} \times N_{\phi} \times 8 \text{ bytes} = 51 \times 128 \times 256 \times 512 \times 8 \approx \boxed{855.64\text{MB}(\Delta t)^{-1}}$

	20M	50M
• Total (100ms or 20000M):		
	Low resolution	Low resolution
	$\approx 213.91 \text{ GB}$	$\approx 85.56 \text{ GB}$
	High resolution	High resolution
	$\approx 855.64 \text{ GB}$	$\approx 346.58 \text{ GB}$

GRMHD butterfly diagrams

VARIABLES

Primitives: `bcc1`, `bcc2`, `bcc3`, `velx`, `vely`, `velz`

Gauge: `z4c_alpha`, `z4c_betax`, `z4c_betay`, `z4c_betaz`

Metric: `adm_gxx`, `adm_gxy`, `adm_gxz`, `adm_gyy`, `adm_gyz`, `adm_gzz`

Total: $N_{\text{vars}} = 16$

OUTPUT TYPE

```
<output4>
file_type = sph
variable = ... # see variable list
dt        = 20.0
radius    = 400.0 # radii = 300.0 400.0 ...
```

SETUP

Output time: 10M or 20M (the latter might already be too large)

Resolution: $(\theta, \phi)_{\text{low}} = (128, 256)$ or $(\theta, \phi)_{\text{high}} = (256, 512)$

Radii: Somewhere in the disk (between 50M and 300M)

DISK USAGE

- Low resolution - $N_{\text{vars}} \times N_{\theta} \times N_{\phi} \times 8 \text{ bytes} = 16 \times 128 \times 256 \times 8 \approx 4.19\text{MB}(r\Delta t)^{-1}$
- High resolution - $N_{\text{vars}} \times N_{\theta} \times N_{\phi} \times 8 \text{ bytes} = 16 \times 256 \times 512 \times 8 \approx 16.78\text{MB}(r\Delta t)^{-1}$

	10M	20M
• Total (100ms or 20000M):		
	Low resolution	High resolution
	$\approx 8.38 \text{ GBr}^{-1}$	$\approx 33.56 \text{ GBr}^{-1}$
	$\approx 4.19 \text{ GBr}^{-1}$	$\approx 16.78 \text{ GBr}^{-1}$

NOTE To save disk-usage, this can be paired with the output from **Ejecta** and also **Neutrino luminosity and average energy density**. The computation is performed via $b_{\bar{\phi}} = \frac{x b_y - y b_x}{\sqrt{x^2 + y^2}}$, where b_x and b_y are the magnetic field components along the x- and y-direction in the fluid frame, and $\bar{\phi}$ denotes the average in the azimuthal direction.

Waveforms

VARIABLES None.

OUTPUT TYPE A block like this is needed inside of the `<z4c>` block:

```
nrad_wave_extraction = 5
extraction_radius_0   = 100
extraction_radius_1   = 200
extraction_radius_2   = 300
extraction_radius_3   = 400
extraction_radius_4   = 500
waveform_dt           = 4.0
```

This will generate two files (real/imag) for each radius with all the modes up to $l = 8$ by default.

SETUP

Output time: 10M is a good minimum, but 20M might also be acceptable Radii: a wide range between 100M and 1000M is recommended for proper waveform extraction

DISK USAGE

- Bytes per line - $N_{\text{byte,column}} \times N_{\text{columns}} = \underbrace{21}_{\text{worstcase}} \times \underbrace{77}_{\text{for 1 passive scalar}} + \underbrace{13}_{\text{t-column}} \approx 1708 \text{ Bytes per file}$

Num. radii /dt	5M	10M	20M
• Total (100ms or 20000M):			
5	$\approx 68.32 \text{ MB}$	$\approx 34.16 \text{ MB}$	$\approx 17.08 \text{ MB}$
10	$\approx 136.64 \text{ MB}$	$\approx 68.32 \text{ MB}$	$\approx 34.16 \text{ MB}$

Kinetic/magnetic energy evolution

■ **VARIABLES** None.

■ **OUTPUT TYPE**

```
<output5>
file_type   = hst
dt          = 20.0
data_format = %20.15e
```

For this analysis we require the file `<jobname>.mhd.hst`.

■ **SETUP**

Output time: 5M is a good minimum, but can also be increased to 10M or 20M, or lowered to 1M

■ **DISK USAGE**

- Bytes per line - $N_{\text{byte,column}} \times N_{\text{columns}} = \underbrace{22}_{21 \text{ if unsigned}} \times \underbrace{14}_{\text{for 1 passive scalar}} \approx 294 \text{ Bytes}$

- Total (100ms or 20000M):

	1M	5M	10M	20M
Usage	$\approx 5.88 \text{ MB}$	$\approx 1.18 \text{ MB}$	$\approx 0.588 \text{ MB}$	$\approx 0.294 \text{ MB}$

User history output

■ **VARIABLES** None.

■ **OUTPUT TYPE**

```
<output6>
file_type   = hst
dt          = 20.0
data_format = %20.15e
```

For this analysis we require the file `<jobname>.user.hst`. Note: the parameter `user_hist` inside of `<problem>` is needed for this output.

■ **SETUP**

Output time: 5M is a good minimum, but can also be increased to 10M or 20M, or lowered to 1M

■ **DISK USAGE**

- Bytes per line - $N_{\text{byte,column}} \times N_{\text{columns}} = \underbrace{22}_{21 \text{ if unsigned}} \times 4 \approx 88 \text{ Bytes}$

- Total (100ms or 20000M):

	1M	5M	10M	20M
Usage	$\approx 1.76 \text{ MB}$	$\approx 0.352 \text{ MB}$	$\approx 0.176 \text{ MB}$	$\approx 0.088 \text{ MB}$

Spectra (MHD)

■ **VARIABLES**

Primitives: `bcc1`, `bcc2`, `bcc3`, `velx`, `vely`, `velz`

Total: $N_{\text{vars}} = 6$

■ **OUTPUT TYPE**

```
<output7>
file_type = bin
variable  = ... # see variable list
dt        = 20.0
```

■ SETUP

Output time: 50M or 100M

Start: post-merger phase

■ DISK USAGE

- Usage per MeshBlock (32^3) - $N_{\text{vars}} \times N_{\text{points per MB}} = 6 \times 32^3 \times 8 \text{ bytes} \approx$ 1.57MB per MeshBlock
- Usage per MeshBlock (48^3) - $N_{\text{vars}} \times N_{\text{points per MB}} = 6 \times 48^3 \times 8 \text{ bytes} \approx$ 5.31MB per MeshBlock
- Usage per MeshBlock (64^3) - $N_{\text{vars}} \times N_{\text{points per MB}} = 6 \times 64^3 \times 8 \text{ bytes} \approx$ 12.58MB per MeshBlock

• Total (100ms or 20000M):		50M	100M
	$N_{\text{p.p.MB}} = 32^3; N_{\text{MB tot.}} = 1000$	$\approx 628 \text{ GB}$	$\approx 314 \text{ GB}$
	$N_{\text{p.p.MB}} = 48^3; N_{\text{MB tot.}} = 1000$	$\approx 2.124 \text{ TB}$	$\approx 1.062 \text{ TB}$
	$N_{\text{p.p.MB}} = 64^3; N_{\text{MB tot.}} = 1000$	$\approx 5.032 \text{ TB}$	$\approx 2.516 \text{ TB}$